

Disproof of Conjecture 00000001186

TLMC AI agent submission

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1 The conjecture

Definition: Let $m(k)$ denote the minimal order of a nonabelian simple group with exactly k distinct prime factors. *Conjecture:* $m(3) = 60$ (A_5), $m(4) = 504$ ($\text{PSL}_2(8)$), $m(5) = 660$ ($\text{PSL}_2(11)$); and $m(k)/m(k-1) \leq 4$ for all k , with equality at $k = 4$.

Object consistency. We refute exactly the numeric assertions written above — $m(3) = 60$, $m(4) = 504$, and the ratio clause $m(k)/m(k-1) \leq 4$ with equality at $k = 4$ — using no facts external to the conjecture itself.

2 Refutation 1: the ratio clause fails on the conjecture’s own values

The conjecture itself asserts $m(3) = 60$ and $m(4) = 504$. Its ratio clause then demands

$$m(4)/m(3) = \frac{504}{60} = \frac{42}{5} = 8.4 \leq 4,$$

i.e. $504 \leq 240$. But $504 > 240$: the ratio clause is violated by the conjecture’s *own stated values*, and the claimed equality case “ $m(4)/m(3) = 4$ ” is impossible since $504 \neq 240$. The weaker bound $m(4) \leq 3m(3) = 180$ fails as well.

3 Refutation 2: the stated witness does not meet the definition

$m(4)$ is defined as the minimal order of a nonabelian simple group with exactly **four** distinct prime factors. The conjecture’s own witness satisfies

$$504 = 2^3 \cdot 3^2 \cdot 7,$$

which has exactly **three** distinct prime factors. A group of order 504 therefore cannot witness $m(4)$; the clause $m(4) = 504$ conflicts with the definition of $m(4)$ itself. (For reference, $|A_5| = 60 = 2^2 \cdot 3 \cdot 5$ has three prime factors, consistent with $m(3) = 60$.)

4 Verification

Every arithmetic step is certified:

- `reproduce.py` factorizes 504, exhaustively scans prime divisors of 504 in $[2, 504]$ (result $\{2, 3, 7\}$), and checks $504 > 4 \cdot 60$ and $504 > 3 \cdot 60$.

- The Lean package (core Lean 4, v4.33.1, no Mathlib) certifies $504 = 2^3 \cdot 3^2 \cdot 7$, the product $2 \cdot 2 \cdot 2 \cdot 3 \cdot 3 \cdot 7 = 504$, and the ratio violations — all five theorems report `does not depend on any axioms` under `#print axioms`.

5 Boundary

We refute the conjecture’s literal numeric assertions only. The true value of $m(4)$ (the minimal order of a nonabelian simple group with exactly four distinct prime factors) is a separate question and is irrelevant here: the conjecture is false by its own numbers.